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# Devoir 2
# CSI2520 Paradigmes de Programmation
**Hiver 2026**

# *Assignment 2*
# CSI2120 Programming Paradigms
**Winter 2026**

## 6% - 18 points

**Question 1. (8 points)**

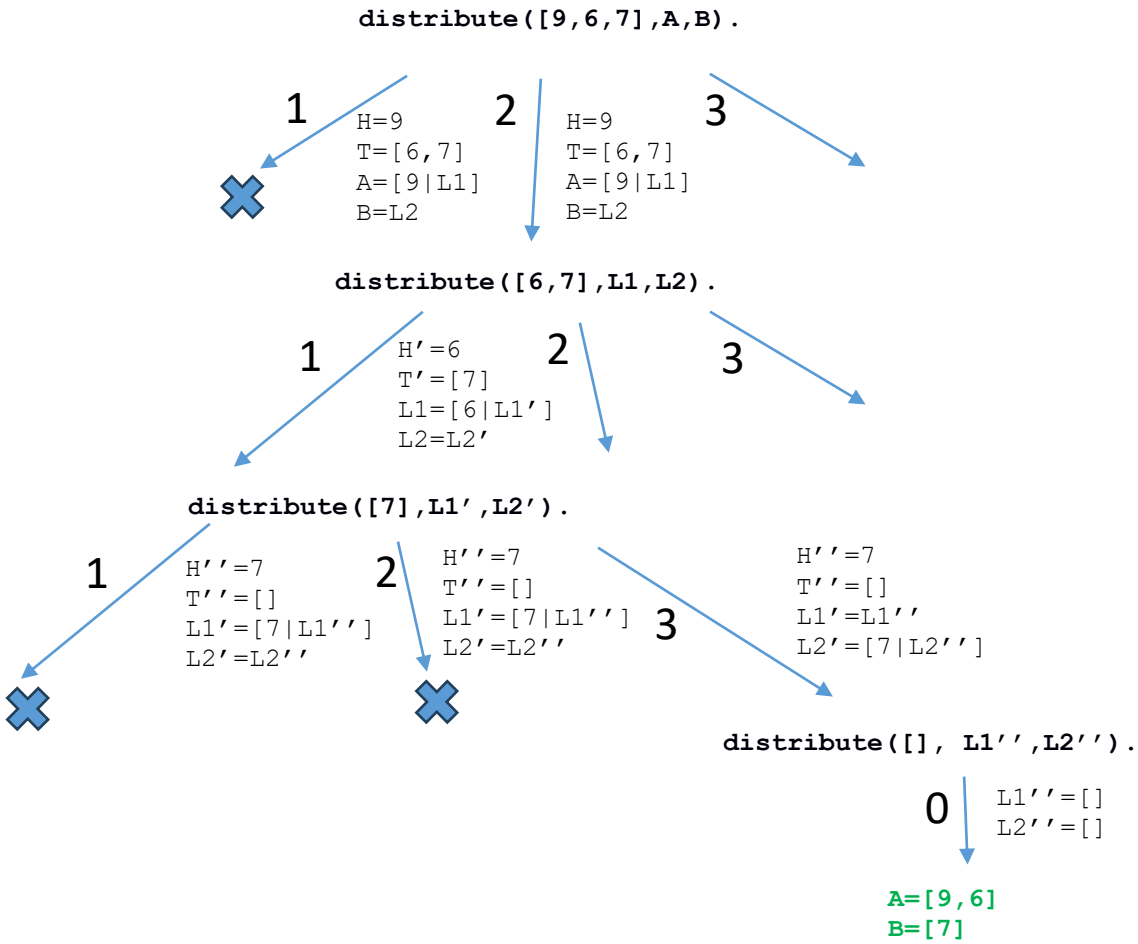
Soit le programme Prolog suivant :
*Consider the following Prolog program:*

```
divisible(X,N):- 0 is X mod N.        %0
distribute([], [], []).
distribute([H|T], [H|L1], L2) :-      %1
    divisible(H,2),
    distribute(T, L1, L2).
distribute([H|T], [H|L1], L2) :-      %2
    divisible(H,3),
    distribute(T, L1, L2).
distribute([H|T], L1, [H|L2]) :-      %3
    distribute(T, L1, L2).
```

a) [4 points] Dessiner l'arbre de résolution Prolog complet pour la requête ci-dessous. Bien montrer les solutions trouvées.
*Draw the complete Prolog search tree for the following query. Show all solutions found.*

```
?- distribute([9,6,7],A,B).
```

La page suivante montre l'arbre partiel menant à la première solution.
*The next page shows the partial tree for the first solution found.*


distribute([9,6,7],A,B).
1
H=9
T=[6,7]
A=[9|L1]
B=L2
2
H=9
T=[6,7]
A=[9|L1]
B=L2
3
distribute([6,7],L1,L2).
1
H'=6
T'=[7]
L1=[6|L1']
L2=L2'
2
3
distribute([7],L1',L2').
1
H''=7
T''=[]
L1'=[7|L1'']
L2'=L2''
2
H''=7
T''=[]
L1'=[7|L1'']
L2'=L2''
3
H''=7
T''=[]
L1'=L1''
L2'=[7|L2'']
distribute([], L1'',L2'').
0
L1''=[]
L2''=[]
A=[9,6]
B=[7]

b) [2+2 points] Pour chacune des coupes ajoutées ci-dessous, reproduire votre arbre complet et montrer quelles seront les branches élaguées par l'introduction de ces coupes (donner la liste des solutions trouvées)

*For each of the added cut below, reproduce your tree and show which branches will be pruned by these cuts (give the list of the solutions found)*

i)

```
divisible(X,N):- 0 is X mod N.        %0
distribute([], [], []).
distribute([H|T], [H|L1], L2) :-      %1
    divisible(H,2),!,
    distribute(T, L1, L2).
distribute([H|T], [H|L1], L2) :-      %2
    divisible(H,3),
    distribute(T, L1, L2).
distribute([H|T], L1, [H|L2]) :-      %3
    distribute(T, L1, L2).
```

ii)

```
divisible(X,N):- 0 is X mod N.        %0
distribute([], [], []).
distribute([H|T], [H|L1], L2) :-      %1
    divisible(H,2),!,
    distribute(T, L1, L2).
distribute([H|T], [H|L1], L2) :-      %2
    divisible(H,3),!,
    distribute(T, L1, L2).
distribute([H|T], L1, [H|L2]) :-      %3
    distribute(T, L1, L2).
```

**Question 2. (10 points)**

Soit la base de faits Prolog ci-dessous :
*Consider the following Prolog database:*

```
room(room101).
room(room202).
room(auditorium).
day(monday).
day(tuesday).
day(wednesday).
day(thursday).
day(friday).
work_hours(8,18). % events cannot start before, or end after
% event(event_name, day, start_hour, end_hour, room_name)
event(meeting1, monday, 9, 11, room101).
event(meeting2, monday, 14, 17, room101).
event(workshop, monday, 8, 17, room202).
event(cleanup, monday, 8, 18, auditorium).
event(conference, tuesday, 14, 16, room101).
event(board, tuesday, 9, 12, room202).
event(activity, tuesday, 10, 17, auditorium).
```

a) [2 points] Écrire le prédicat `busy_room(Day, Start, End, Room)` permettant de déterminer si une salle est occupée pour la période mentionnée.
*Write the predicate* `busy_room(Day, Start, End, Room)` *that determines if a room is occupied for the period specified.*

```
?- busy_room(monday, 13, 15, room101).
true .

?- busy_room(tuesday, 8, 11, R).
R = room202 ;
R = auditorium.

?- busy_room(D, 13, 15, room101).
D = monday ;
D = tuesday ;
false.
```

b) [2 points] Écrire le prédicat `free_room(Day, Start, End, Room)` permettant de déterminer si une salle est libre pour la période mentionnée.
*Write the predicate* `free_room(Day, Start, End, Room)` *that determines if a room is available for the period specified.*

```
?- free_room(D, 13, 15, room101).
D = wednesday ;
D = thursday ;
D = friday.

?- free_room(tuesday, 8, 11, R).
R = room101 ;
false.
```

c) [2 points] Écrire le prédicat `find_available_room(Day, Duration, Room, Start)` permettant de trouver une salle disponible pour un événement ayant une certaine durée.
*Write the predicate* `find_available_room(Day, Duration, Room, Start)` *that finds a room for an event of a specified duration.*

```
?- find_available_room(monday,2,Room,Start).
Room = room101,
Start = 11 ;
Room = room101,
Start = 12 ;
false.

?- find_available_room(Day,9,auditorium,Start).
Day = wednesday,
Start = 8 ;
Day = thursday,
Start = 8 ;
Day = friday,
Start = 8 ;
Day = wednesday,
Start = 9 ;
Day = thursday,
Start = 9 ;
Day = friday,
Start = 9 ;
false.
```

d) [2 points] Écrire le prédicat `hours_occupied(Room, Day, Hours)` permettant de calculer le nombre d'heures pendant laquelle une salle est occupée pour un jour donné.
*Write the predicate* `hours_occupied(Room, Day, Hours)` *that calculates the number of hours a room is occupied for a given day.*

```
?- hours_occupied(room101,J,H).
J = monday,
H = 5 ;
J = tuesday,
H = 2 ;
J = wednesday,
H = 0 ;
J = thursday,
H = 0 ;
J = friday,
H = 0.
```

e) [2 points] Écrire le prédicat `hours_occupied(Room, Hours)` permettant de calculer le nombre total d'heures qu'une salle est occupée dans la semaine.
*Write the predicate* `hours_occupied(Room, Hours)` *that calculates the total number of hours a room is occupied for the week.*

```
?- hours_occupied(room101,H).
H = 5.
```